

CASTERTON, Australia

WMO: 958250

Lat: 37.5831S

Lon: 141.3339E

Elev: 131

StdP: 99.76

Time Zone: 10.00 (AUS)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.0	2.2	-0.4	3.7	10.4	0.7	4.0	8.0	10.6	12.1	9.7	11.2	1.0	60	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	35.4	18.4	32.7	18.1	30.2	17.6	20.5	28.6	19.6	27.8	18.7	27.1	5.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	22.1	17.1	12.4	21.1	16.0	11.6	20.4	59.5	28.0	56.3	27.4	53.4	27.1	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.5	7.7	-1.1	40.5	1.1	2.4	-1.8	42.2	-2.5	43.6	-3.1	45.0	-3.9	46.7		
(5)			-1.3	22.6	1.1	1.2	-2.1	23.5	-2.8	24.2	-3.4	24.9	-4.2	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	19.6	19.4	17.7	14.4	11.6	9.2	8.9	9.5	10.8	12.8	15.2	17.1	(6)
(7)		DBStd	5.06	4.37	4.01	4.14	3.27	2.63	2.00	2.02	2.29	2.62	3.55	4.03	4.09	(7)
(8)		HDD10.0	164	0	0	0	2	12	38	46	37	19	9	2	0	(8)
(9)		HDD18.3	1895	33	32	63	125	210	273	292	273	227	177	114	76	(9)
(10)		CDD10.0	1562	299	264	240	135	61	14	12	22	42	97	157	219	(10)
(11)		CDD18.3	251	73	62	45	8	1	0	0	0	0	6	20	36	(11)
(12)		CDH23.3	3274	1011	703	526	89	1	0	0	0	2	92	286	565	(12)
(13)		CDH26.7	1488	528	322	229	22	0	0	0	0	0	23	102	262	(13)
(14)	Wind	WSAvg	3.9	4.1	4.1	3.9	3.3	3.4	3.1	4.0	4.1	4.1	4.1	4.2	(14)	
(15)	Precipitation	PrecAvg	659	29	24	37	44	60	72	87	86	75	60	47	38	(15)
(16)		PrecMax	899	89	95	119	130	158	158	187	165	143	147	126	138	(16)
(17)		PrecMin	370	5	0	4	5	8	13	34	13	28	13	10	4	(17)
(18)		PrecStd	120	19	21	27	29	32	31	34	33	29	32	23	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	36.2	36.6	29.3	21.6	16.2	15.8	19.9	22.6	29.7	32.6	37.3	(19)
(20)			MCWB	19.7	18.8	17.9	16.4	14.3	12.6	12.8	13.0	15.3	16.4	18.4	18.5	(20)
(21)		2%	DB	36.4	33.6	32.0	25.7	19.2	14.7	14.1	17.4	20.0	25.9	29.8	33.2	(21)
(22)			MCWB	18.7	18.6	17.8	15.4	13.5	11.9	11.6	12.1	13.8	15.7	17.9	17.2	(22)
(23)		5%	DB	32.5	30.9	28.9	23.1	17.4	13.7	13.1	15.0	17.8	22.8	26.8	29.4	(23)
(24)			MCWB	18.1	18.3	17.1	14.4	13.4	11.6	10.9	11.4	12.8	14.7	17.3	17.3	(24)
(25)		10%	DB	28.9	27.7	25.9	20.6	16.0	12.8	12.2	13.5	16.0	20.1	23.5	26.0	(25)
(26)			MCWB	17.5	17.6	16.4	14.0	12.8	11.0	10.3	10.6	12.1	13.9	16.3	16.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.9	20.4	17.7	15.7	13.8	13.8	14.3	16.1	18.9	21.3	21.1	(27)
(28)			MCDWB	31.5	30.6	26.2	24.0	18.4	14.8	14.5	17.3	19.7	25.8	28.9	28.9	(28)
(29)		2%	WB	20.4	20.0	19.0	16.5	14.8	12.7	12.5	12.8	14.6	16.6	19.6	19.2	(29)
(30)			MCDWB	29.1	29.3	27.8	21.9	17.3	13.8	13.4	15.9	18.4	22.8	26.5	27.7	(30)
(31)		5%	WB	19.4	19.0	18.1	15.7	14.0	12.0	11.5	11.7	13.4	15.5	18.2	18.1	(31)
(32)			MCDWB	28.3	27.2	26.0	20.4	16.3	13.1	12.5	14.4	17.0	21.6	24.8	26.5	(32)
(33)		10%	WB	18.3	18.1	17.2	14.8	13.3	11.3	10.7	11.0	12.3	14.4	16.7	17.1	(33)
(34)			MCDWB	27.5	26.2	24.3	19.0	15.4	12.5	11.9	13.1	15.6	19.3	22.9	25.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	13.2	12.3	10.4	7.7	7.3	6.6	7.6	9.2	11.3	12.4	14.0	(35)
(36)			MCDBR	20.6	18.3	17.7	15.2	9.3	8.4	7.1	9.9	11.7	16.5	18.0	20.5	(36)
(37)			MCWBR	6.3	5.8	5.8	6.5	5.4	6.2	5.3	6.1	6.9	8.0	8.0	6.9	(37)
(38)		5% WB	MCDWB	16.0	14.9	14.2	11.5	7.9	7.3	6.6	8.5	10.6	14.1	15.3	15.9	(38)
(39)			MCWBR	6.4	5.9	5.6	5.8	5.1	5.8	5.3	5.7	6.8	7.7	7.9	6.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.333	0.321	0.310	0.304	0.298	0.298	0.312	0.333	0.336	0.335	0.335	(40)	
(41)		taud	2.503	2.533	2.564	2.578	2.571	2.576	2.572	2.519	2.449	2.447	2.483	2.503	(41)	
(42)		Ebn at Noon	994	977	950	899	844	819	842	882	919	962	992	1003	(42)	
(43)		Edh at Noon	112	105	94	81	70	65	69	84	103	113	114	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.31	6.34	4.94	3.43	2.28	1.85	2.02	2.87	3.99	5.31	6.33	7.09	(44)	
(45)		RadStd	0.44	0.31	0.23	0.23	0.12	0.14	0.12	0.15	0.31	0.45	0.44	0.45	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		+0.39	N/A	N/A	+0.90	N/A	N/A	N/A	-96	+132	+39

Nomenclature: See separate page